Mozes Jacobs

425-273-5242 | mozesjacobs@g.harvard.edu | mozesjacobs.github.io

Education

Harvard University — Cambridge, MA
Ph.D. in Computer Science | Advisor: Demba Ba | Kempner Graduate Fellow — 2023 – 2028 (Expected)
Computer Vision, Representation Learning, Interpretability, Foundation Models

University of Washington — Seattle, WA
B.S. in Computer Science | Summa Cum Laude | GPA: 3.99 / 4.00 — 2018 – 2022

Publications & Preprints

[1] **A Phenomenology of Attention Sinks: Two Algorithms, Two Solutions** — **Under Review**
Fesser L.*, **Jacobs M.***, Fel T.*, Keller TA., Kakade S.

[2] **Block-Recurrent Dynamics in ViTs** — **ICLR 2026**
Jacobs M.*, Fel T.*, Hakim R.*, Brondetta A., Ba D., Keller TA.

[3] **Traveling Waves Integrate Spatial Information Through Time** — **CCN 2025 (oral)**
Jacobs M., Budzinski RC., Muller L., Ba D., Keller TA.

[4] **HyperSINDY: Deep Generative Modeling of Nonlinear Stochastic Governing Equations** — **Preprint 2023**
Jacobs M., Brunton BW., Brunton SL., Kutz JN., Raut

Research Experience

Kempner Institute | **Harvard University** | *Graduate Research Fellow* — 2023 – Present

- **Block-Recurrent Foundation Models:** Showed foundation models (e.g., DINOv2) can be distilled into recurrent surrogates that recover **96%** of ImageNet-1k accuracy with only **2 recurrent blocks**, enabling principled model compression.
- **Dynamical Interpretability:** Characterized ViT depth as a dynamical flow into low-dimensional attractors, revealing token-specific self-correcting trajectories and low-rank late-layer updates.
- **Neuro-Inspired Architectures:** Designed convolutional recurrent models using traveling waves for spatial integration, outperforming U-Nets on semantic segmentation.

Allen Institute | **AI Institute in Dynamic Systems** | *Shanahan Foundation Predoctoral Fellow* — 2022 – 2023

- **HyperSINDy:** Built a deep generative framework combining SINDy with amortized variational inference for uncertainty-aware discovery of stochastic governing equations. *(Advisors: R. Raut, J.N. Kutz, S. Brunton)*

Neural Systems Lab | **University of Washington** | | *Undergraduate Research Assistant* — 2019 – 2022

- **Predictive Coding:** Proposed *Gradient Origin Predictive Coding*, a video generation framework unifying cortical predictive coding with Bayesian deep learning for next-frame prediction.

Noble Lab | **University of Washington** | *Undergraduate Research Assistant* — 2019 – 2022

- **Genomic Algorithms:** Optimized *PASTIS*, a Poisson-based 3D chromatin structure inference algorithm, reducing memory overhead and runtime.

Teaching Experience

Harvard University | *Teaching Fellow, CS 2790R: Human-Computer Interaction* — Fall 2025

- Facilitated student learning in graduate-level HCI concepts under Prof. Elena Glassman, holding weekly office hours and providing comprehensive feedback on project designs and implementations.

Harvard University | *Teaching Fellow, APMTH 231: Decision Theory* — Spring 2026

- Evaluated advanced coursework and provided technical feedback on core decision theory and statistical methodologies under the instruction of Prof. Demba Ba.

Invited Talks

[1] **Traveling Waves Integrate Spatial Information Through Time**
Neural Computations: Dynamics Across Space, Time, and Task — **CCN 2025 (Aug 2025)**

[2] **Can Your Neurons Hear the Shape of an Object?**
Frontiers in NeuroAI Symposium — **Kempner Institute (June 2025)**

Awards & Fellowships

Kempner Institute Graduate Fellowship — 2023 – Present
Shanahan Foundation Predoctoral Fellowship — 2022
Burkhardt Family Endowed Scholarship — 2020, 2021
Gary A. Kildall Endowed Scholarship — 2019
Boeing Scholarship — 2018

Technical Skills

Languages & Frameworks: Python, PyTorch, Java